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8.2.1 Basics of descent methods

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▶ 2.0x 🔊 🗑️ 📺 “

the following idea. What if we have a current guess as to where the solution is? This is our point, $\hat{x}$. Let's say that we have had some kind of method where at the current step we are at $x^{(k)}$. Okay? What we can do? Now ideally we would say we want to go directly from there to the solution. But, well that requires solving the linear system. And that's the whole point of the method. Right? So what if instead we pick some

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Reading assignment

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Read Unit 8.2.1 of the notes. [[LINK](#)]

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Homework 8.2.1.1

1/1 point (graded)
The cost of an iterative method is a combination of how many iterations it takes to convergence and the cost per iteration. For loops in the algorithm in

```
Given :  $A, b, x^{(0)}$   
 $r^{(0)} := b - Ax^{(0)}$   
 $k := 0$   
while  $r^{(k)} \neq 0$   
     $p^{(k)} :=$  next direction  
     $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$   
     $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$   
     $r^{(k+1)} := b - Ax^{(k+1)}$   
     $k := k + 1$   
endwhile
```

count the number of matrix-vector multiplications, dot products, and "axpy" operations (not counting the cost of determining the next descent direction).

2

✓

matrix-vector multiplies (mvmults)

Answer: 2

2

✓

dot products (dots)

Answer: 2

1

✓

"axpy"s

Answer: 1

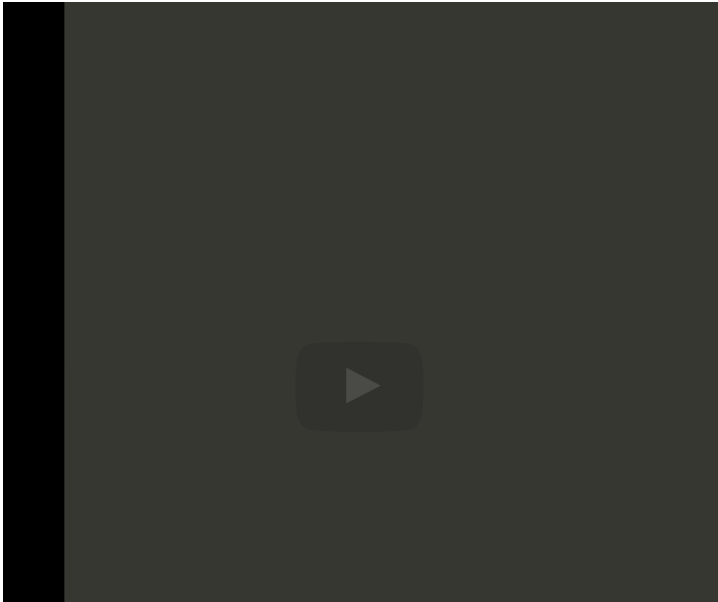
Solution:

For a complete solution, see the [notes](#).

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at all vectors of the form $x^{(k)}$ plus alpha times that search direction. And let's minimize that." In other words, let's pick the alpha that minimizes this. And let's call that α_k . Then our next approximation to the solution is just our current approximation plus α_k times the search direction. And then we also

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compute the residual.
And at that point

we're ready to go
back to the top. And
we just keep doing
this.

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Homework 8.2.1.2

1/1 point (graded)
For loops in the algorithm in

Given : $A, b, x^{(0)}$
 $r^{(0)} := b - Ax^{(0)}$
 $k := 0$
while $r^{(k)} \neq 0$
 $p^{(k)} :=$ next direction
 $q^{(k)} := Ap^{(k)}$
 $\alpha_k := \frac{p^{(k)T}r^{(k)}}{p^{(k)T}q^{(k)}}$
 $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$
 $r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$
 $k := k + 1$
endwhile

count the number of matrix-vector multiplications, dot products, and "axpy" operations (not counting the cost of determining the next descent direction).

1

✓

matrix-vector multiplies (mvmults)

Answer: 1

2

✓

dot products (dots)

Answer: 2

2

✓

"axpy"s

Answer: 2

Solution:

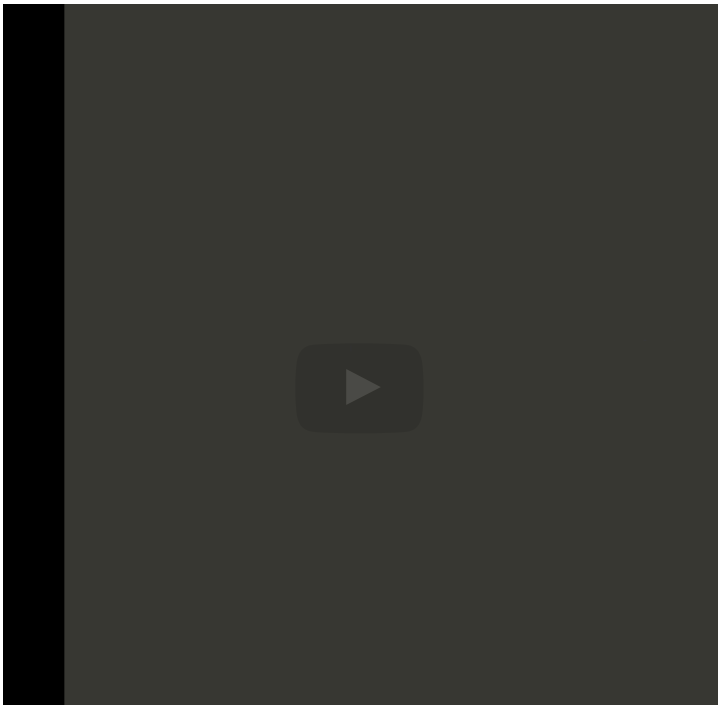
For a complete solution, see the [notes](#).

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So simple calculus showed that this alpha k then is



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α_k then is actually equal to the dot product of the current search direction with the current residual, divided by this quantity right here: $p^{(k)T} A p^{(k)}$. And notice that since A is symmetric positive definite as long as our search direction is not the zero vector, **this actually is nonzero. As a matter of fact it's positive.**



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